

# FC7xxx FLS User Manual

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Rev.0.6

# Revision History

| Revision | Date       | Changes                                               |
|----------|------------|-------------------------------------------------------|
| 0.1      | 2023/07/14 | Initial release for MCAL V0.1.0                       |
| 0.3      | 2023/10/20 | Updated for MCAL V0.3.0                               |
| 0.6      | 2023/03/29 | Updated for MCAL V0.6.0<br>- Added support for FC7240 |

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# Chapter 1 FLS Introduction

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## 1.1 Requirement Tracing

The design of this module follows the specifications of the Flash driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR\_SWS\_FlashDriver*.

## 1.2 Hardware Summary

The flash driver provides services for reading, writing and erasing flash memory and a configuration interface for setting / resetting the write / erase protection if supported by the underlying hardware.

In application mode of the ECU, the flash driver is only to be used by the Flash EEPROM emulation module for writing data. It is not intended to write program code to flash memory in application mode. This shall be done in boot mode which is out of scope of AUTOSAR.

# Chapter 2 Software Design

## 2.1 Rejected Requirements

| Rejected Requirement 1 <b>SWS_Fls_00088</b> |                                                                                                                                                    |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | The functional requirements and the functional scope are the same for both internal and external drivers. Hence the API is semantically identical. |
| <b>Rejection Reason</b>                     | Disable config external drivers, only use internal flash.                                                                                          |

| Rejected Requirement 2 <b>SWS_Fls_00107</b> |                                                                                 |
|---------------------------------------------|---------------------------------------------------------------------------------|
| <b>Description</b>                          | file structure.                                                                 |
| <b>Rejection Reason</b>                     | Internal flash does not use interrupt, so it does not include Flash_Irq.c file. |

| Rejected Requirement 3 <b>SWS_Fls_00319</b> |                                                                                                                        |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | The transient fault code FLS_E_UNEXPECTED_FLASH_ID shall be reported when the expected flash ID is not matched.        |
| <b>Rejection Reason</b>                     | FLS_E_UNEXPECTED_FLASH_ID is applicable only for external Fls driver.h, here only support internal flash, so not used. |

| Rejected Requirement 4 <b>SWS_Fls_00144</b> |                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | During the initialization of the external flash driver, the FLS module shall check the hardware ID of the external flash device against the corresponding published parameter. If a hardware ID mismatch occurs, the FLS module shall report the error code FLS_E_UNEXPECTED_FLASH_ID to the Default Error Tracer (DET), set the FLS module status to FLS_E_UNINIT and shall not initialize itself. |
| <b>Rejection Reason</b>                     | Only support internal flash, so not check and report.                                                                                                                                                                                                                                                                                                                                               |

| Rejected Requirement 5 <b>SWS_Fls_00137</b> |                                                                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | The FLS module's implementer shall place the code of the flash access routines into a separate C-module Fls_ac.c. |
| <b>Rejection Reason</b>                     | Flash access code is fixed in ROM, and just call the corresponding API.                                           |

| Rejected Requirement 6 <b>SWS_Fls_00213</b> |                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | The FLS module's main processing routine shall access the flash access code routines by means of the respective function pointer contained in the FLS module's configuration set (post-compile parameters) regardless whether the flash access code routines have been loaded to RAM or whether they can be executed directly from (flash) ROM. |
| <b>Rejection Reason</b>                     | Flash access code is fixed in ROM and just call the corresponding API.                                                                                                                                                                                                                                                                          |

| Rejected Requirement 7 <b>SWS_Fls_0048</b> |  |
|--------------------------------------------|--|
|--------------------------------------------|--|

|                         |                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | If supported by hardware, the function Fls_Init shall set the flash memory erase/write protection as provided in the configuration set. |
| <b>Rejection Reason</b> | not implemented                                                                                                                         |

**Rejected Requirement 8 SWS\_Fls\_00221**

|                         |                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | The job of the function Fls_Erase shall erase a flash memory block starting from FlsBaseAddress + TargetAddress of size Length. Note: Length will be rounded up to the next full sector boundary since only complete flash sectors can be erased. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                                                                         |

**Rejected Requirement 9 SWS\_Fls\_00020**

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | If development error detection for the module Fls is enabled: the function Fls_Erase shall check that the erase start address (FlsBaseAddress + TargetAddress) is aligned to a flash sector boundary and that it lies within the specified lower and upper flash address boundaries. If this check fails, the function Fls_Erase shall reject the erase request, raise the development error FLS_E_PARAM_ADDRESS and return with E_NOT_OK. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                                                                                                                                                                                                                                                                  |

**Rejected Requirement 10 SWS\_Fls\_00145**

|                         |                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | If possible, e.g. with interrupt controlled implementations, the FLS module shall start the first round of the erase job directly within the function Fls_Erase to reduce overall runtime. |
| <b>Rejection Reason</b> | not use interrupt, so not do this                                                                                                                                                          |

**Rejected Requirement 11 SWS\_Fls\_00226**

|                         |                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | The job of the function Fls_Write shall program a flash memory block with data provided via SourceAddressPtr starting from FlsBaseAddress + TargetAddress of size Length. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                 |

**Rejected Requirement 12 SWS\_Fls\_00026**

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | If development error detection for the module Fls is enabled: the function Fls_Write shall check that the write start address (FlsBaseAddress + TargetAddress) is aligned to a flash page boundary and that it lies within the specified lower and upper flash address boundaries. If this check fails, the function Fls_Write shall reject the write request, raise the development error FLS_E_PARAM_ADDRESS and return with E_NOT_OK. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                                                                                                                                                                                                                                                                |

**Rejected Requirement 13 SWS\_Fls\_00239**

|                         |                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | The read job of the function Fls_Read shall copy a continuous flash memory block starting from FlsBaseAddress + SourceAddress of size Length to the buffer pointed to by TargetAddressPtr. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                  |

**Rejected Requirement 14 SWS\_Fls\_00097**

|                         |                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | If development error detection for the module Fls is enabled: the function Fls_Read shall check that the read start address (FlsBaseAddress + SourceAddress) lies within the specified lower and upper flash address boundaries. If this check fails, the function Fls_Read shall reject the read job, raise development error FLS_E_PARAM_ADDRESS and return with E_NOT_OK. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                                                                                                                                                                                                    |

**Rejected Requirement 15 SWS\_Fls\_00244**

|                         |                                                                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | The job of the function Fls_Compare shall compare a continuous flash memory block starting from FlsBaseAddress + SourceAddress of size Length with the buffer pointed to by TargetAddressPtr. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                     |

**Rejected Requirement 16 SWS\_Fls\_00150**

|                         |                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | If development error detection for the module Fls is enabled: the function Fls_Compare shall check that the compare start address (FlsBaseAddress + SourceAddress) lies within the specified lower and upper flash address boundaries. If this check fails, the function Fls_Compare shall reject the compare job, raise the development error FLS_E_PARAM_ADDRESS and return with E_NOT_OK. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                                                                                                                                                                                                                    |

**Rejected Requirement 17 SWS\_Fls\_00379**

|                         |                                                                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | The verification job of the function Fls_BlankCheck shall check, that the continuous flash memory area starting from FlsBaseAddress + TargetAddress of size Length is erased. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                     |

**Rejected Requirement 18 SWS\_Fls\_00380**

|                         |                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | If development error detection for the module FLS is enabled; the function Fls_BlankCheck shall check that the verification start address (FlsBaseAddress + TargetAddress) lies within the specified lower and upper flash address boundaries. If this check fails, the function Fls_BlankCheck shall reject the verification job, raise the development error FLS_E_PARAM_ADDRESS and return with E_NOT_OK. |
| <b>Rejection Reason</b> | FlsBaseAddress not used, unclear purpose.                                                                                                                                                                                                                                                                                                                                                                    |

**Rejected Requirement 19 SWS\_Fls\_00193**

|                    |                                                                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Depending on implementation, callback routines provided and/or invoked by the FLS module may be called on interrupt level. The module providing those routines has therefore to make sure that their runtime is reasonably short, i.e. since callbacks may be propagated upward through several software layers. |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



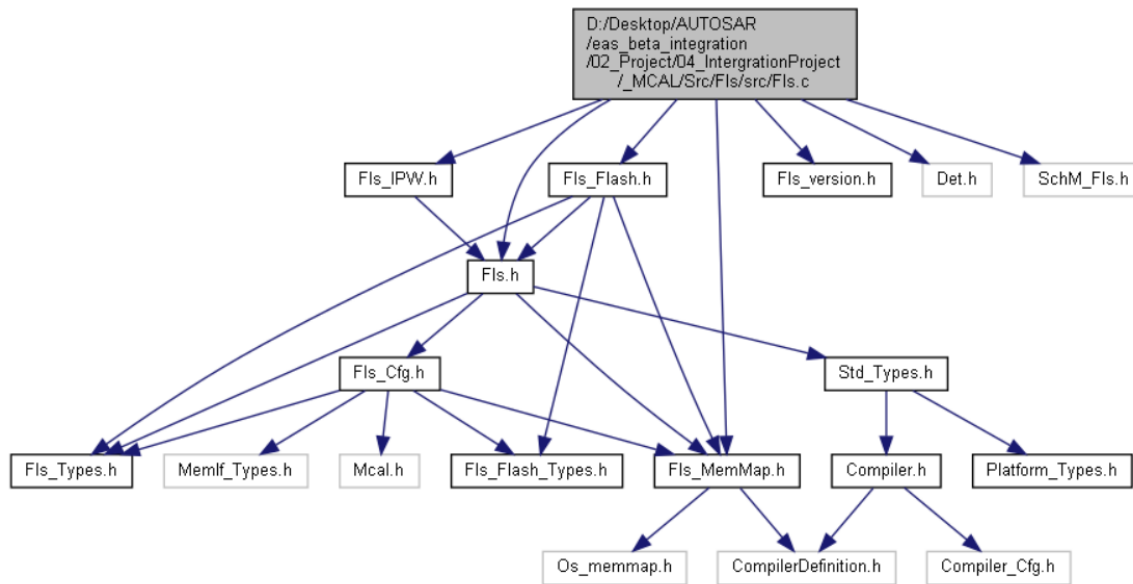
|                         |                                    |
|-------------------------|------------------------------------|
| <b>Rejection Reason</b> | Not necessary and not implemented. |
|-------------------------|------------------------------------|

|                                                                               |                                                                                       |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Rejected Requirement 20 <span style="color: red;">SWS_Fls_00233</span></b> |                                                                                       |
| <b>Description</b>                                                            | The FLS module's implementer shall locate the interrupt service routine in Fls_Irq.c. |
| <b>Rejection Reason</b>                                                       | Hardware does not support Interrupt, so not have FLS_Irq.c                            |

|                                                                               |                                                                             |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Rejected Requirement 21 <span style="color: red;">SWS_Fls_00354</span></b> |                                                                             |
| <b>Description</b>                                                            | The type declaration for Fls_ConfigType shall be located in the file Fls.h. |
| <b>Rejection Reason</b>                                                       | Fls_ConfigType is included into Fls.h from Fls_Cfg.h.                       |

| Rejected Requirement 22 ECUC_Fls_00317 |                                  |                                                                  |    |              |
|----------------------------------------|----------------------------------|------------------------------------------------------------------|----|--------------|
| Description                            | SWS Item                         | ECUC_Fls_00317                                                   |    |              |
|                                        | Name                             | FlsSpiReference                                                  |    |              |
|                                        | Parent Container                 | FlsExternalDriver                                                |    |              |
|                                        | Description                      | Reference to SPI sequence (required for external Flash drivers). |    |              |
|                                        | Multiplicity                     | 1..*                                                             |    |              |
|                                        | Type                             | Symbolic name reference to [ SpiSequence ]                       |    |              |
|                                        | Post-Build Variant Multiplicity  | false                                                            |    |              |
|                                        | Post-Build Variant Value         | false                                                            |    |              |
|                                        | Multiplicity Configuration Class | Pre-compile time                                                 | X  | All Variants |
|                                        |                                  | Link time                                                        | -- |              |
|                                        |                                  | Post-build time                                                  | -- |              |
|                                        | Value Configuration Class        | Pre-compile time                                                 | X  | All Variants |
|                                        |                                  | Link time                                                        | -- |              |
|                                        |                                  | Post-build time                                                  | -- |              |
|                                        | Scope / Dependency               | scope: local                                                     |    |              |
| Rejection Reason                       | Not used, so not implemented.    |                                                                  |    |              |

## 2.2 File Structure



## 2.3 Macros

### 2.3.1 Macros in Fls.h

- #define **FLS\_E\_PARAM\_CONFIG** 0x01U
- #define **FLS\_E\_PARAM\_ADDRESS** 0x02U
- #define **FLS\_E\_PARAM\_LENGTH** 0x03U
- #define **FLS\_E\_PARAM\_DATA** 0x04U  
API service called with wrong parameter.
- #define **FLS\_E\_UNINIT** 0x05U  
API service called without module initialization.
- #define **FLS\_E\_BUSY** 0x06U  
API service called while driver was still busy.
- #define **FLS\_E\_ERASE\_FAILED** 0x01U  
Flash erase failed (HW).
- #define **FLS\_E\_WRITE\_FAILED** 0x02U  
Flash write failed (HW).
- #define **FLS\_E\_READ\_FAILED** 0x03U  
Flash read failed (HW).

- **#define FLS\_E\_COMPARE\_FAILED 0x04U**  
Flash compare failed (HW).
- **#define FLS\_E\_UNEXPECTED\_FLASH\_ID 0x05U**  
Expected hardware ID not matched.
- **#define FLS\_E\_VERIFY\_ERASE\_FAILED 0x07U**  
Erase verification (blank check) failed.
- **#define FLS\_E\_VERIFY\_WRITE\_FAILED 0x08U**  
Write verification (compare) failed.
- **#define FLS\_E\_TIMEOUT 0x09U**  
Timeout exceeded.
- **#define FLS\_E\_PARAM\_POINTER 0x0AU**  
API service called with NULL pointer.
- **#define FLS\_INIT\_ID 0x00U**  
Init service ID.
- **#define FLS\_ERASE\_ID 0x01U**  
Erase service ID.
- **#define FLS\_WRITE\_ID 0x02U**  
Write service ID.
- **#define FLS\_CANCEL\_ID 0x03U**  
Cancel service ID.
- **#define FLS\_GETSTATUS\_ID 0x04U**  
GetStatus service ID.
- **#define FLS\_GETJOBRESULT\_ID 0x05U**  
GetJobResult service ID.
- **#define FLS\_MAINFUNCTION\_ID 0x06U**  
MainFunction service ID.
- **#define FLS\_READ\_ID 0x07U**  
Read service ID.
- **#define FLS\_COMPARE\_ID 0x08U**  
Compare service ID.

- #define **FLS\_SETMODE\_ID** 0x09U  
SetMode service ID.
- #define **FLS\_GETVERSIONINFO\_ID** 0x10U  
GetVersionInfo service ID.
- #define **FLS\_BLANK\_CHECK\_ID** 0x11U  
BlankCheck service ID.

### 2.3.2 Macros in Fls\_version.h

- #define **FLS\_AR\_RELEASE\_MAJOR\_VERSION** 4
- #define **FLS\_AR\_RELEASE\_MINOR\_VERSION** 6
- #define **FLS\_AR\_RELEASE\_REVISION\_VERSION** 0
- #define **FLS\_MODULE\_ID** 92
- #define **FLS\_SW\_MAJOR\_VERSION** 0
- #define **FLS\_SW\_MINOR\_VERSION** 6
- #define **FLS\_SW\_PATCH\_VERSION** 0
- #define **FLS\_VENDOR\_ID** 174

### 2.3.3 Macros in FLS\_Cfg.h

- #define **FLS\_CFG\_AR\_RELEASE\_MAJOR\_VERSION** 4
- #define **FLS\_CFG\_AR\_RELEASE\_MINOR\_VERSION** 6
- #define **FLS\_CFG\_AR\_RELEASE\_REVISION\_VERSION** 0
- #define **FLS\_CFG\_SW\_MAJOR\_VERSION** 0
- #define **FLS\_CFG\_SW\_MINOR\_VERSION** 6
- #define **FLS\_CFG\_SW\_PATCH\_VERSION** 0
- #define **FLS\_CFG\_VENDOR\_ID** 174

- #define **FLS\_AC\_LOAD\_ON\_JOB\_START** (STD\_OFF)  
Flash access code loaded on job start / unloaded on job end or error.
- #define **FLS\_CANCEL\_API** (STD\_ON)  
Compile switch to enable and disable the Fls\_Cancel function.
- #define **FLS\_COMPARE\_API** (STD\_ON)  
Compile switch to enable and disable the Fls\_Compare function.
- #define **FLS\_BLANK\_CHECK\_API** (STD\_ON)  
Compile switch to enable and disable the Fls\_BlankCheck function.
- #define **FLS\_DEV\_ERROR\_DETECT** (STD\_ON)  
Pre-processor switch to enable and disable development error detection.
- #define **FLS\_GET\_JOB\_RESULT\_API** (STD\_ON)  
Compile switch to enable and disable the Fls\_GetJobResult function.
- #define **FLS\_GET\_STATUS\_API** (STD\_ON)  
Compile switch to enable and disable the Fls\_GetStatus function.
- #define **FLS\_SET\_MODE\_API** (STD\_ON)  
Compile switch to enable and disable the Fls\_SetMode function.
- #define **FLS\_USE\_INTERRUPTS** (STD\_OFF)  
Pre-processor switch to enable / disable the API to use interrupts for erase and write jobs.
- #define **FLS\_VERSION\_INFO\_API** (STD\_ON)  
Pre-processor switch to enable / disable the API to read out the modules version information.
- #define **FLS\_WRITE\_VERIFY\_CHECK** (STD\_ON)  
Pre-processor switch to enable / disable the write verify check.
- #define **FLS\_ERASE\_VERIFY\_CHECK** (STD\_ON)  
Pre-processor switch to enable / disable the erase blank check.
- #define **FLS\_TIMEOUT\_HANDLING** (STD\_ON)  
Timeout handling enabled.
- #define **FLS\_AC\_SIZE\_ERASE** (0U)  
Number of bytes in RAM needed for the erase flash access code.

- #define **FLS\_AC\_SIZE\_WRITE** (0U)  
Number of bytes in RAM needed for the write flash access code.
- #define **FLS\_ERASED\_VALUE** (0xffffffffU)  
The contents of an erased flash memory cell.
- #define **FLS\_DRIVER\_INDEX** (0U)  
Fls Driver Index.
- #define **D\_FLASH\_BASE\_ADDR** (0x4000000U)  
The flash memory start address.
- #define **D\_FLASH\_PAGE\_SIZE** (0x10U)  
Size of one page of this sector.
- #define **D\_FLASH\_SECTOR\_SIZE** (0x2000U)  
Size of one sector.
- #define **D\_FLASH\_MAX\_WRITE\_SIZE** (0x80U)  
Size of once max write.

## 2.4 Enums

### 2.4.1 Enumerations in Fls\_Types.h

#### 2.4.1.1 Fls\_JobType

| Enumeration | Fls_JobType                                         |                                                   |
|-------------|-----------------------------------------------------|---------------------------------------------------|
| Description | Type of job currently executed by Fls_MainFunction. |                                                   |
| Values      | Value                                               | Description                                       |
|             | FLS_JOB_ERASE                                       | Erase one or more complete flash sectors.         |
|             | FLS_JOB_WRITE                                       | Write one or more complete flash pages.           |
|             | FLS_JOB_READ                                        | Read one or more bytes from flash memory.         |
|             | FLS_JOB_COMPAR                                      | Compare data buffer with content of flash memory. |
|             | FLS_JOB_BLANK_CHECK                                 | Check content of erased flash memory area.        |

#### 2.4.1.2 Fls\_LLDJobType

| Enumeration | Fls_LLDJobType                                         |                                                             |
|-------------|--------------------------------------------------------|-------------------------------------------------------------|
| Description | Type of job currently executed by Fls_LLDMainFunction. |                                                             |
| Values      | Value                                                  | Description                                                 |
|             | FLASH_JOB_NONE                                         | No job executed by Fls_LLDMainFunction.                     |
|             | FLASH_JOB_ERASE                                        | Erase one flash sector.                                     |
|             | FLASH_JOB_ERASE_TEMP                                   | Complete erase and start an interleaved erase flash sector. |
|             | FLASH_JOB_WRITE                                        | Write one or more complete flash pages.                     |

|  |                             |                                    |
|--|-----------------------------|------------------------------------|
|  | FLASH_JOB_ERASE_BLANK_CHECK | Erase blank check of flash sector. |
|--|-----------------------------|------------------------------------|

### 2.4.1.3 Fls\_LLDReturnTypes

|                    |                                      |                                                         |
|--------------------|--------------------------------------|---------------------------------------------------------|
| <b>Enumeration</b> | Fls_LLDReturnTypes                   |                                                         |
| <b>Description</b> | Result of low-level flash operation. |                                                         |
| <b>Values</b>      | <b>Value</b>                         | <b>Description</b>                                      |
|                    | FLASH_E_OK                           | Operation succeeded.                                    |
|                    | FLASH_E_FAILED                       | Operation failed due to hardware error.                 |
|                    | FLASH_E_BLOCK_INCONSISTENT           | Data buffer doesn't match with content of flash memory. |
|                    | FLASH_E_PENDING                      | Operation is pending.                                   |
|                    | FLASH_E_PARTITION_ERR                | FlexNVM partition ratio error.                          |

## 2.5 Typedefs

### 2.5.1 Typedefs in Fls\_Types.h

- typedef **uint32 Fls\_SectorIndexType**  
Logical sector index.
- typedef **uint16 Fls\_CrcType**  
CRC computed over config set.
- typedef **uint32 Fls\_AddressType**  
Address offset from the configured flash base address to access a certain flash memory area.
- typedef **uint32 Fls\_LengthType**  
Number of bytes to read, write, erase, compare.
- typedef **uint32 Fls\_SectorCountType**  
Number of configured sectors.
- typedef **void (\*Fls\_JobEndNotificationPtrType)(void);**  
Pointer type of Fls\_JobEndNotification function.
- typedef **void (\*Fls\_JobErrorNotificationPtrType)(void);**  
Pointer type of Fls\_JobErrorNotification function.

## 2.6 Structures

### 2.6.1 Fls\_ConfigType

|                    |                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structure</b>   | Fls_ConfigType                                                                                                                                  |
| <b>Description</b> | A pointer to such a structure is provided to the flash driver initialization routine for configuration of the driver and flash memory hardware. |
| <b>Diagram</b>     | N/A                                                                                                                                             |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Fields</b> | <ul style="list-style-type: none"> <li>• <code>Fls_JobEndNotificationPtrType jobEndNotificationPtr</code><br/>Pointer to job end notification function.</li> <li>• <code>Fls_JobErrorNotificationPtrType jobErrorNotificationPtr</code><br/>Pointer to job error notification function.</li> <li>• <code>MemIf_ModeType eDefaultMode</code><br/>Default FLS device mode after initialization (MEMIF_MODE_FAST, MEMIF_MODE_SLOW).</li> <li>• <code>Fls_LengthType u32MaxReadFastMode</code><br/>Max number of bytes to read in one cycle of <code>Fls_MainFunction</code> (fast mode).</li> <li>• <code>Fls_LengthType u32MaxReadNormalMode</code><br/>Max number of bytes to read in one cycle of <code>Fls_MainFunction</code> (normal mode).</li> <li>• <code>Fls_LengthType u32MaxWriteFastMode</code><br/>Max number of bytes to write in one cycle of <code>Fls_MainFunction</code> (fast mode).</li> <li>• <code>Fls_LengthType u32MaxWriteNormalMode</code><br/>Max number of bytes to write in one cycle of <code>Fls_MainFunction</code> (normal mode).</li> <li>• <code>Fls_SectorCountType u32BlockCount</code><br/>Number of configured logical sectors.</li> <li>• <code>const Fls_AddressType (*paBlockEndAddr)[]</code><br/>Pointer to array containing last logical address of each configured sector.</li> <li>• <code>const uint16 (*paNumberOfSectorCross)[]</code><br/>Pointer to array containing cross number of HW sector.</li> <li>• <code>const Fls_AddressType (*paBlockStartAddr)[]</code><br/>Pointer to array containing start logical address of each configured sector.</li> <li>• <code>const Fls_Flash_InternalSectorInfoType * const (*pBlockList)[]</code><br/>Pointer to array containing physical sector ID of each configured sector.</li> <li>• <code>Fls_CrcType u16ConfigCrc</code><br/>FLS Config Set CRC checksum.</li> </ul> |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 2.7 API Functions

### 2.7.1 Functions in Fls.h

#### 2.7.1.1 `void Fls_Init(const Fls_ConfigType * pConfigPtr)`

|                    |                                                               |                                            |
|--------------------|---------------------------------------------------------------|--------------------------------------------|
| <b>Function</b>    | <code>void Fls_Init(const Fls_ConfigType * pConfigPtr)</code> |                                            |
| <b>Description</b> | The function initializes Fls module.                          |                                            |
| <b>Diagram</b>     | N/A                                                           |                                            |
| <b>Parameters</b>  | <b>Parameter</b>                                              | <b>Description</b>                         |
|                    | <code>pConfigPtr</code>                                       | Pointer to flash driver configuration set. |
| <b>Returns</b>     | None                                                          |                                            |

#### 2.7.1.2 `Std_ReturnType Fls_Erase(Fls_AddressType u32TargetAddress, Fls_LengthType u32Length)`

|                    |                                                                                                   |                    |
|--------------------|---------------------------------------------------------------------------------------------------|--------------------|
| <b>Function</b>    | <code>Std_ReturnType Fls_Erase(Fls_AddressType u32TargetAddress, Fls_LengthType u32Length)</code> |                    |
| <b>Description</b> | Erases flash sector(s).                                                                           |                    |
| <b>Diagram</b>     | N/A                                                                                               |                    |
| <b>Parameters</b>  | <b>Parameter</b>                                                                                  | <b>Description</b> |
|                    |                                                                                                   |                    |



|                |                  |                                 |
|----------------|------------------|---------------------------------|
|                | u32TargetAddress | Target address in flash memory. |
|                | u32Length        | Number of bytes to erase.       |
| <b>Returns</b> | Std_ReturnType:  |                                 |

### 2.7.1.3 Std\_ReturnType Fls\_Write(Fls\_AddressType u32TargetAddress, const uint8\* pSourceAddressPtr, Fls\_LengthType u32Length )

|                    |                                                                                                                       |                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Function</b>    | Std_ReturnType Fls_Write(Fls_AddressType u32TargetAddress, const uint8* pSourceAddressPtr, Fls_LengthType u32Length ) |                                                                                                     |
| <b>Description</b> | Writes one or more complete flash pages.                                                                              |                                                                                                     |
| <b>Diagram</b>     | NA                                                                                                                    |                                                                                                     |
| <b>Parameters</b>  | <b>Parameter</b>                                                                                                      | <b>Description</b>                                                                                  |
|                    | TargetAddress                                                                                                         | Target address in flash memory. This address offset will be added to the flash memory base address. |
|                    | SourceAddressPtr                                                                                                      | Pointer to source data buffer.                                                                      |
|                    | Length                                                                                                                | Number of bytes to write.                                                                           |
| <b>Returns</b>     | Std_ReturnType                                                                                                        |                                                                                                     |

### 2.7.1.4 void Fls\_Cancel( oid)

|                    |                         |
|--------------------|-------------------------|
| <b>Function</b>    | void Fls_Cancel( oid)   |
| <b>Description</b> | Cancels an ongoing job. |
| <b>Diagram</b>     | N/A                     |
| <b>Parameters</b>  | None                    |
| <b>Returns</b>     | None                    |

### 2.7.1.5 MemIf\_StatusType Fls\_GetStatus(void)

|                    |                                      |
|--------------------|--------------------------------------|
| <b>Function</b>    | MemIf_StatusType Fls_GetStatus(void) |
| <b>Description</b> | Returns the driver state.            |
| <b>Diagram</b>     | N/A                                  |
| <b>Parameters</b>  | None                                 |
| <b>Returns</b>     | MemIf_StatusType                     |

### 2.7.1.6 MemIf\_JobResultType Fls\_GetJobResult (void)

|                    |                                             |
|--------------------|---------------------------------------------|
| <b>Function</b>    | MemIf_JobResultType Fls_GetJobResult (void) |
| <b>Description</b> | Returns the result of the last job.         |
| <b>Diagram</b>     | N/A                                         |
| <b>Parameters</b>  | None                                        |
| <b>Returns</b>     | MemIf_JobResultType                         |

### 2.7.1.7 Std\_ReturnType Fls\_Read( Fls\_AddressType u32SourceAddress, uint8\* pTargetAddressPtr, Fls\_LengthType u32Length )

|                    |                                                                                                                 |                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Function</b>    | Std_ReturnType Fls_Read( Fls_AddressType u32SourceAddress, uint8* pTargetAddressPtr, Fls_LengthType u32Length ) |                                                                                                     |
| <b>Description</b> | Reads from flash memory.                                                                                        |                                                                                                     |
| <b>Diagram</b>     | N/A                                                                                                             |                                                                                                     |
| <b>Parameters</b>  | <b>Parameter</b>                                                                                                | <b>Description</b>                                                                                  |
|                    | u32SourceAddress                                                                                                | Source address in flash memory. This address offset will be added to the flash memory base address. |
|                    | u32Length                                                                                                       | Number of bytes to read.                                                                            |
| <b>Returns</b>     | Std_ReturnType                                                                                                  |                                                                                                     |

### 2.7.1.8 Std\_ReturnType Fls\_Compare( Fls\_AddressType u32SourceAddress, const uint8\* pTargetAddressPtr, Fls\_LengthType u32Length)

|                    |                                                                                                                         |                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Function</b>    | Std_ReturnType Fls_Compare( Fls_AddressType u32SourceAddress, const uint8* pTargetAddressPtr, Fls_LengthType u32Length) |                                                                                                     |
| <b>Description</b> | Compares the contents of an area of flash memory with that of an application data                                       |                                                                                                     |
| <b>Diagram</b>     | N/A                                                                                                                     |                                                                                                     |
| <b>Parameters</b>  | <b>Parameter</b>                                                                                                        | <b>Description</b>                                                                                  |
|                    | u32SourceAddress                                                                                                        | Source address in flash memory. This address offset will be added to the flash memory base address. |
|                    | u32Length                                                                                                               | Number of bytes to compare.                                                                         |
| <b>Returns</b>     | N/A                                                                                                                     |                                                                                                     |

### 2.7.1.9 void Fls\_SetMode( MemIf\_ModeType eMode )

|                    |                                          |                    |
|--------------------|------------------------------------------|--------------------|
| <b>Function</b>    | void Fls_SetMode( MemIf_ModeType eMode ) |                    |
| <b>Description</b> | Sets the flash driver's operation mode.  |                    |
| <b>Diagram</b>     | N/A                                      |                    |
| <b>Parameters</b>  | <b>Parameter</b>                         | <b>Description</b> |
|                    | MEMIF_MODE_SLOW                          | Slow read access.  |
|                    | MEMIF_MODE_FAST                          | Fast read access.  |
| <b>Returns</b>     | N/A                                      |                    |

### 2.7.1.10 void Fls\_GetVersionInfo(Std\_VersionInfoType\* pVersioninfoPtr)

|                    |                                                               |                                                                   |
|--------------------|---------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Function</b>    | void Fls_GetVersionInfo(Std_VersionInfoType* pVersioninfoPtr) |                                                                   |
| <b>Description</b> | Returns the version information of this module.               |                                                                   |
| <b>Diagram</b>     | N/A                                                           |                                                                   |
| <b>Parameters</b>  | <b>Parameter</b>                                              | <b>Description</b>                                                |
|                    | pVersioninfoPtr                                               | Pointer to where to store the version information of this module. |
| <b>Returns</b>     | None                                                          |                                                                   |

**2.7.1.11 Std\_ReturnType Fls\_BlankCheck( Fls\_AddressType u32TargetAddress, Fls\_LengthType u32Length )**

|                    |                                                                                                                                                                                                                                                                                             |                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Function</b>    | Std_ReturnType Fls_BlankCheck( Fls_AddressType u32TargetAddress, Fls_LengthType u32Length )                                                                                                                                                                                                 |                                                                       |
| <b>Description</b> | The function Fls_BlankCheck shall verify, whether a given memory area has been erased but not (yet) programmed. The function shall limit the maximum number of checked flash cells per main function cycle to the configured value FlsMaxReadNormalMode or FlsMaxReadFastMode respectively. |                                                                       |
| <b>Diagram</b>     | N/A                                                                                                                                                                                                                                                                                         |                                                                       |
| <b>Parameters</b>  | <b>Parameter</b>                                                                                                                                                                                                                                                                            | <b>Description</b>                                                    |
|                    | u32TargetAddress                                                                                                                                                                                                                                                                            | Address in flash memory from which the blank check should be started. |
|                    | u32Length                                                                                                                                                                                                                                                                                   | Number of bytes to be checked for erase pattern.                      |
| <b>Returns</b>     | Std_ReturnType                                                                                                                                                                                                                                                                              |                                                                       |

**2.7.1.12 void Fls\_MainFunction(void)**

|                    |                                  |
|--------------------|----------------------------------|
| <b>Function</b>    | void Fls_MainFunction(void)      |
| <b>Description</b> | Performs the processing of jobs. |
| <b>Diagram</b>     | N/A                              |
| <b>Parameters</b>  | N/A                              |
| <b>Returns</b>     | N/A                              |

**2.8 Hardware Functions****2.8.1 Functions in Fls\_Flash.h****2.8.1.1 void Fls\_Flash\_Init(void)**

|                      |                            |
|----------------------|----------------------------|
| <b>Function</b>      | void Fls_Flash_Init(void)  |
| <b>Description</b>   | Initialize flash hardware. |
| <b>Parameters</b>    | N/A                        |
| <b>Returns</b>       | N/A                        |
| <b>Referenced By</b> | Fls_Init                   |

**2.8.1.2 void Fls\_Flash\_MainFunctionErase(void)**

|                      |                                        |
|----------------------|----------------------------------------|
| <b>Function</b>      | void Fls_Flash_MainFunctionErase(void) |
| <b>Description</b>   | Process ongoing erase hardware job.    |
| <b>Parameters</b>    | N/A                                    |
| <b>Returns</b>       | N/A                                    |
| <b>Referenced By</b> | Fls_IPW_LLDMainFunction                |

**2.8.1.3 void Fls\_Flash\_MainFunctionWrite(void)**

|                      |                                        |
|----------------------|----------------------------------------|
| <b>Function</b>      | void Fls_Flash_MainFunctionWrite(void) |
| <b>Description</b>   | Enable an interrupt.                   |
| <b>Parameters</b>    | N/A                                    |
| None                 | N/A                                    |
| <b>Referenced By</b> | Fls_IPW_LLDMainFunction                |

## 2.8.1.4 Fls\_Flash\_MainFunctionEraseBlankCheck

|                      |                                             |
|----------------------|---------------------------------------------|
| <b>Function</b>      | Fls_Flash_MainFunctionEraseBlankCheck(void) |
| <b>Description</b>   | Check Erase blank.                          |
| <b>Parameters</b>    | N/A                                         |
| <b>Returns</b>       | N/A                                         |
| <b>Referenced By</b> | Fls_IPW_LLDMainFunction                     |

## 2.8.1.5 void Fls\_Flash\_Cancel(void)

|                      |                                             |
|----------------------|---------------------------------------------|
| <b>Function</b>      | void Fls_Flash_Cancel(void)                 |
| <b>Description</b>   | Cancel ongoing erase or write hardware job. |
| <b>Parameters</b>    | N/A                                         |
| <b>Returns</b>       | N/A                                         |
| <b>Referenced By</b> | Fls_IPW_AbortSuspended                      |

## 2.8.1.6 Fls\_LLDReturnType Fls\_Flash\_Erase(const Fls\_PhysicalSectorType ePhySector)

|                      |                                                                            |                     |
|----------------------|----------------------------------------------------------------------------|---------------------|
| <b>Function</b>      | Fls_LLDReturnType Fls_Flash_Erase(const Fls_PhysicalSectorType ePhySector) |                     |
| <b>Description</b>   | Erase one complete flash sector.                                           |                     |
| <b>Parameters</b>    | <b>Parameter</b>                                                           | <b>Description</b>  |
|                      | ePhySector                                                                 | Physical sector ID. |
| <b>Returns</b>       | Fls_LLDReturnType                                                          |                     |
| <b>Referenced By</b> | Fls_IPW_Erase                                                              |                     |

## 2.8.1.7 Fls\_LLDReturnType Fls\_Flash\_Write(const Fls\_AddressType u32LogicAddr, const Fls\_LengthType u32DataLength, const uint8 \* const pDataPtr)

|                      |                                                                                                                                         |                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Function</b>      | Fls_LLDReturnType Fls_Flash_Write(const Fls_AddressType u32LogicAddr, const Fls_LengthType u32DataLength, const uint8 * const pDataPtr) |                       |
| <b>Description</b>   | Write one or more complete flash pages into given flash sector.                                                                         |                       |
| <b>Parameters</b>    | <b>Parameter</b>                                                                                                                        | <b>Parameter</b>      |
|                      | u32LogicAddr                                                                                                                            | The logic address.    |
|                      | u32DataLength                                                                                                                           | Data length in bytes. |
|                      | pDataPtr                                                                                                                                | Data to be written.   |
| <b>Returns</b>       | Fls_LLDReturnType                                                                                                                       |                       |
| <b>Referenced By</b> | Fls_IPW_Write                                                                                                                           |                       |

## 2.8.1.8 Fls\_LLDReturnType Fls\_Flash\_Read (const Fls\_AddressType u32LogicAddr, Fls\_LengthType u32DataLength, uint8 \* pDataPtr)

|                    |                                                                                                                       |                    |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Function</b>    | Fls_LLDReturnType Fls_Flash_Read (const Fls_AddressType u32LogicAddr, Fls_LengthType u32DataLength, uint8 * pDataPtr) |                    |
| <b>Description</b> | Read one or more bytes from given flash sector.                                                                       |                    |
| <b>Parameters</b>  | <b>Parameter</b>                                                                                                      | <b>Description</b> |

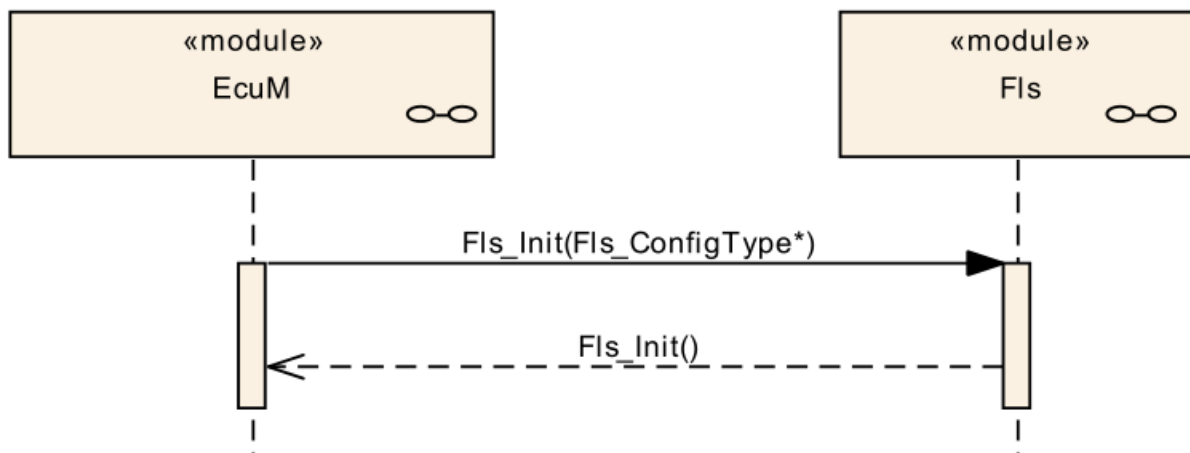
|                      |                    |                                |
|----------------------|--------------------|--------------------------------|
|                      | u32LogicAddr       | The logic address.             |
|                      | u32DataLength      | Data length in bytes.          |
|                      | pDataPtr           | Where read data to be written. |
| <b>Returns</b>       | Fls_LLDReturnTypes |                                |
| <b>Referenced By</b> | Fls_IPW_Read       |                                |

#### 2.8.1.9 Fls\_LLDReturnTypes Fls\_Flash\_Compare (const Fls\_AddressType u32LogicAddr, const Fls\_LengthType u32DataLength, const uint8 \* const pDataPtr)

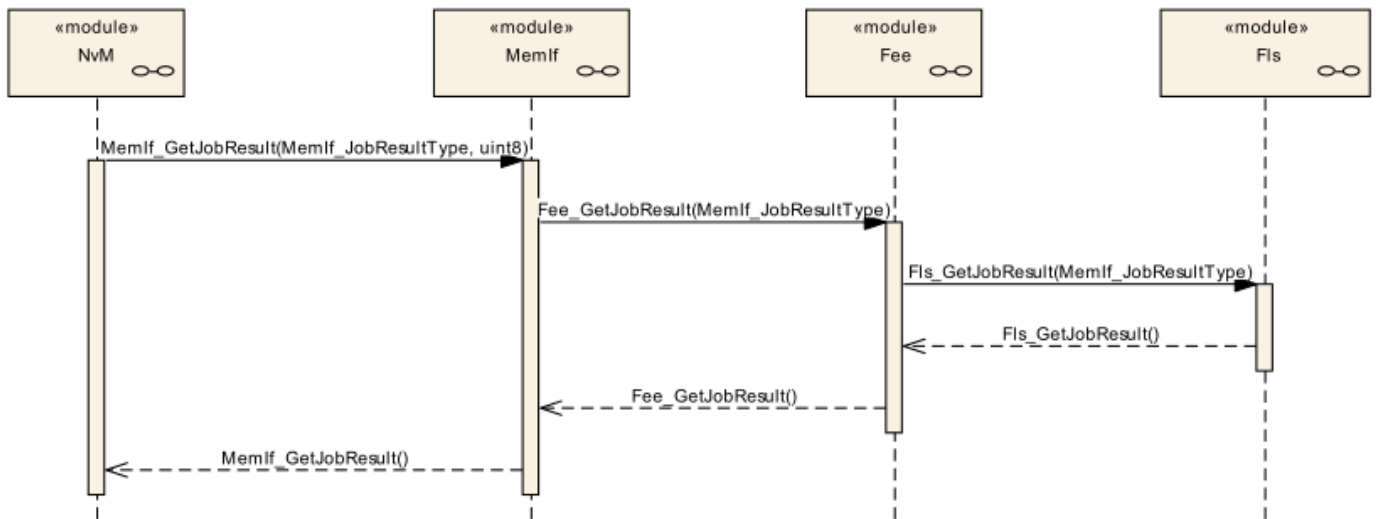
|                      |                                                                                                                                             |                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Function</b>      | Fls_LLDReturnTypes Fls_Flash_Compare (const Fls_AddressType u32LogicAddr, const Fls_LengthType u32DataLength, const uint8 * const pDataPtr) |                       |
| <b>Description</b>   | Compare data buffer with content of given flash sector.                                                                                     |                       |
| <b>Parameters</b>    | <b>Parameter</b>                                                                                                                            | <b>Description</b>    |
|                      | u32LogicAddr                                                                                                                                | The logic address.    |
|                      | u32DataLength                                                                                                                               | Data length in bytes. |
|                      | pDataPtr                                                                                                                                    | Data to be compared.  |
| <b>Returns</b>       | Fls_LLDReturnTypes                                                                                                                          |                       |
| <b>Referenced By</b> | Fls_IPW_Compare                                                                                                                             |                       |

## 2.9 API Sequence Diagram

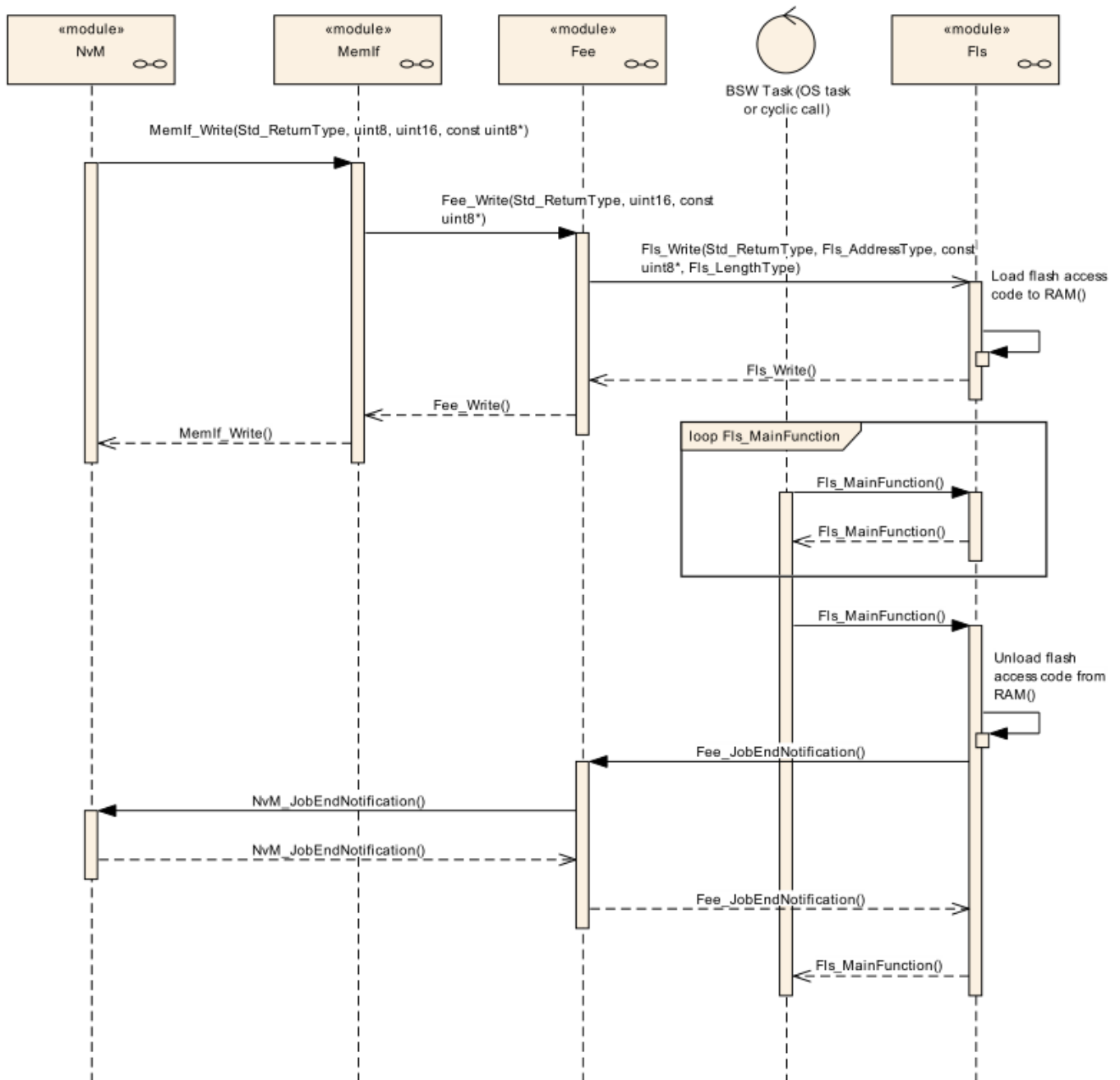
### 2.9.1 Initialization



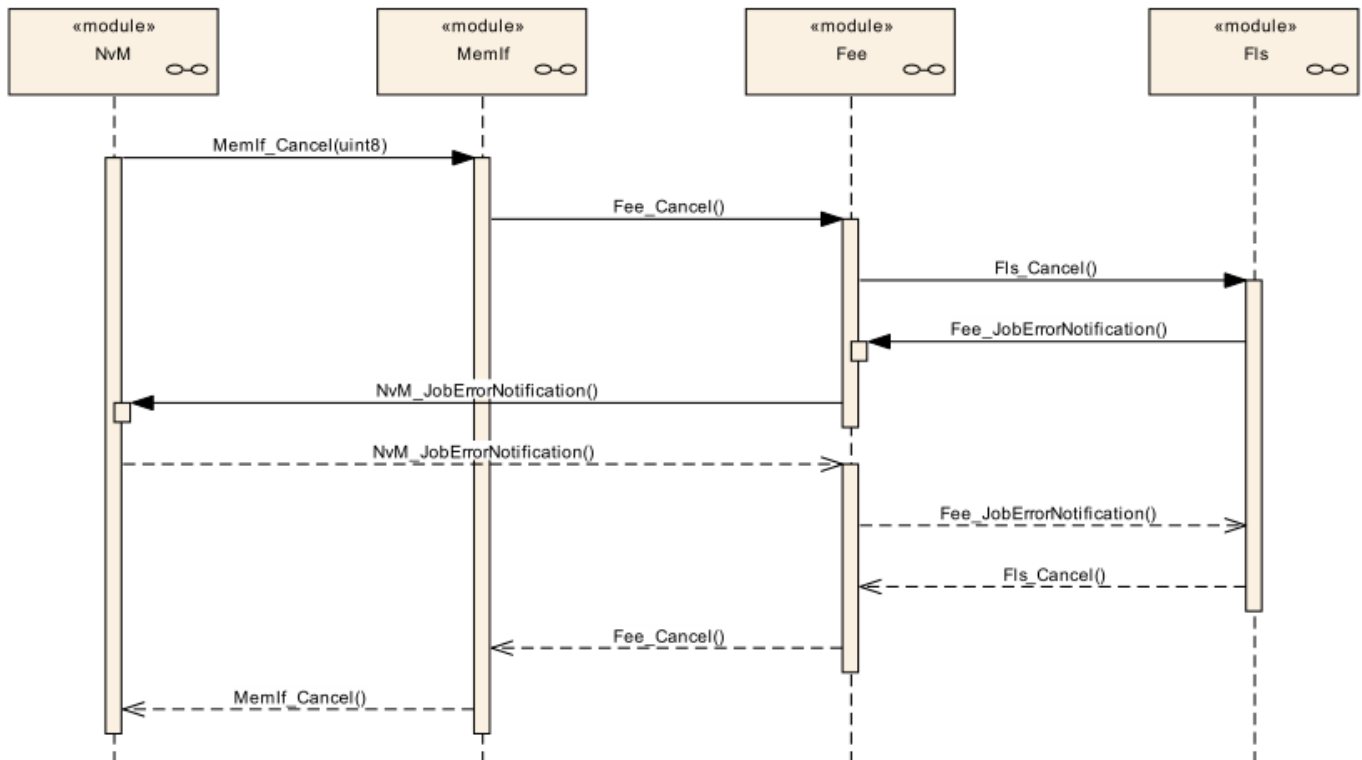
## 2.9.2 Synchronous Functions



## 2.9.3 Asynchronous Functions



## 2.9.4 Canceling a Running Job

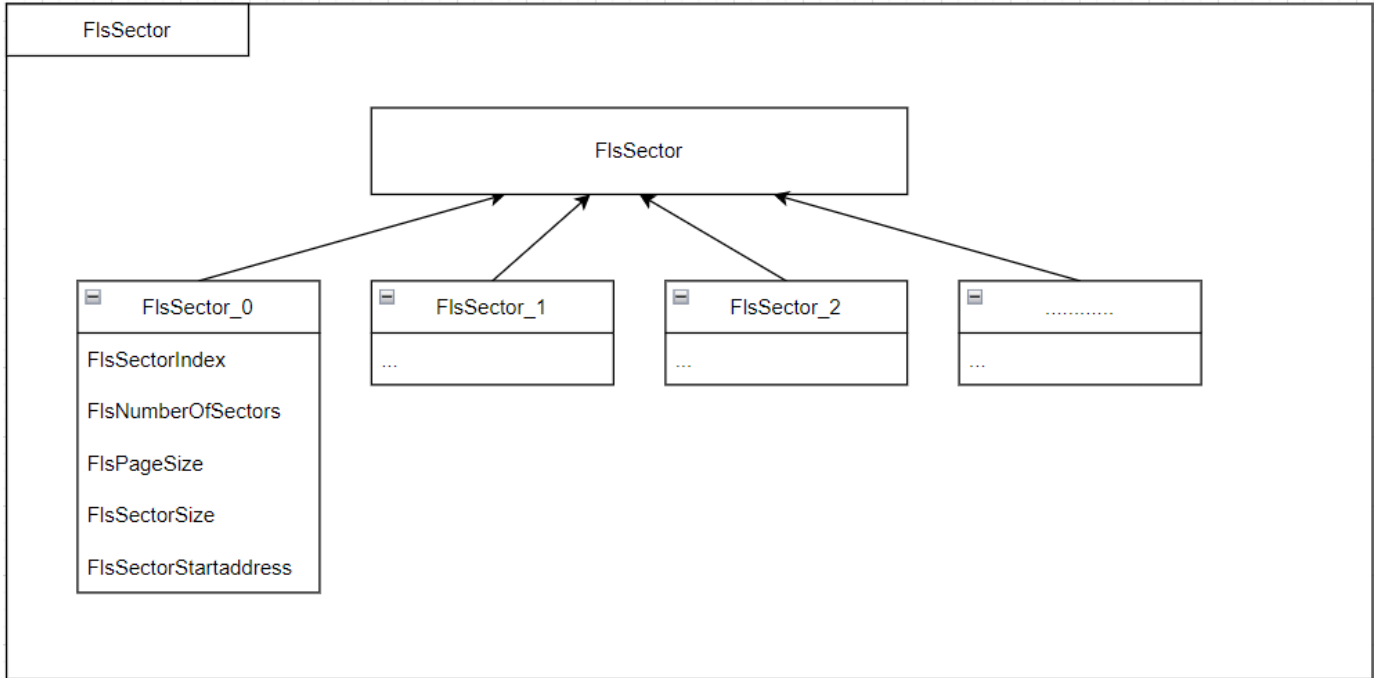




# Chapter 3 Tresos Configuration Items

## 3.1 Container Inclusion Relation

The container inclusion relation is shown below:

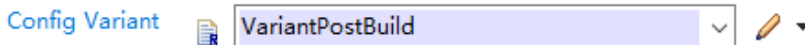


| FlsGeneral                  | FlsConfigSet                |
|-----------------------------|-----------------------------|
| FlsAcLoadOnJobStart         | FlsAcErase                  |
| FlsBaseAddress              | FlsAcWrite                  |
| FlsBlankCheckApi            | FlsCallCycle                |
| FlsCancelApi                | FlsDefaultMode              |
| FlsCompareApi               | FlsJobEndNotification       |
| FlsDevErrorDetect           | FlsJobErrorNotification     |
| FlsDriverIndex              | FlsDriverIndex              |
| FlsEraseVerificationEnabled | FlsEraseVerificationEnabled |
| FlsGetJobResultApi          | FlsMaxReadFastMode          |
| FlsGetStatusApi             | FlsMaxReadNormalMode        |
| FlsSetModeApi               | FlsMaxWriteFastMode         |
| FlsTotalSize                | FlsMaxWriteNormalMode       |
| FlsUseInterrupts            | FlsProtection               |
| FlsVersionInfoApi           |                             |
| FlsWriteVerificationEnabled |                             |

| FlsPublishedInformation | CommonPublishedInformation |
|-------------------------|----------------------------|
| FlsAcLocationErase      | ArReleaseMajorVersion      |
| FlsAcLocationWrite      | ArReleaseMinorVersion      |
| FlsAcSizeErase          | ArReleaseRevisionVersion   |
| FlsAcSizeWrite          | ModuleId                   |
| FlsEraseTime            | SwMajorVersion             |
| FlsErasedValue          | SwMinorVersion             |
| FlsExpectedHwId         | SwPatchVersion             |
| FlsSpecifiedEraseCycles | VendorId                   |
| FlsWriteTime            |                            |

## 3.2 Containers and Variables

### 3.2.1 IMPLEMENTATION\_CONFIG\_VARIANT

|                    |                                                                                    |                                       |
|--------------------|------------------------------------------------------------------------------------|---------------------------------------|
| <b>Variables</b>   | IMPLEMENTATION_CONFIG_VARIANT                                                      |                                       |
| <b>Description</b> | N/A                                                                                |                                       |
| <b>Screenshot</b>  |  |                                       |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b>                          |
|                    | Type                                                                               | Variable: enumeration                 |
|                    | Label                                                                              | Config Variant                        |
|                    | Default                                                                            | VariantPreCompile                     |
|                    | Range                                                                              | VariantPostBuild<br>VariantPreCompile |

### 3.2.2 FlsGeneral

|                    |                                                                                                |              |
|--------------------|------------------------------------------------------------------------------------------------|--------------|
| <b>Container</b>   | FlsConfigSet                                                                                   |              |
| <b>Description</b> | Container for general parameters of the flash driver. These parameters are always pre-compile. |              |
| <b>Screenshot</b>  |            |              |
| <b>Properties</b>  | <b>Property</b>                                                                                | <b>Value</b> |
|                    | Type                                                                                           | Container    |

#### 3.2.2.1 FlsAcLoadOnJobStart

|                    |                                                                                                                                                                            |                   |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Variables</b>   | FlsAcLoadOnJobStart                                                                                                                                                        |                   |
| <b>Description</b> | The flash driver shall load the flash access code to RAM whenever an erase or write job is started and unload (overwrite) it after that job has been finished or canceled. |                   |
| <b>Screenshot</b>  |                                                                                        |                   |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                            | <b>Value</b>      |
|                    | Type                                                                                                                                                                       | Variable: Boolean |

|  |         |                                   |
|--|---------|-----------------------------------|
|  | Label   | Fls Load Access Code On Job Start |
|  | Origin  | AUTOSAR_ECUC                      |
|  | Default | False                             |

### 3.2.2.2 FlsBaseAddress

|                    |                                                                                                                                           |                  |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Variables</b>   | FlsBaseAddress                                                                                                                            |                  |
| <b>Description</b> | The flash memory start address .                                                                                                          |                  |
| <b>Screenshot</b>  | <a href="#">Fls Base Address (0 -&gt; 4294967295)</a>  |                  |
| <b>Properties</b>  | <b>Property</b>                                                                                                                           | <b>Value</b>     |
|                    | Type                                                                                                                                      | Variable:Integer |
|                    | Label                                                                                                                                     | Fls Base Address |
|                    | Origin                                                                                                                                    | AUTOSAR_ECUC     |
|                    | Default                                                                                                                                   | 67108864         |

### 3.2.2.3 FlsBlankCheckApi

|                    |                                                                                                                         |                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Variables</b>   | FlsBlankCheckApi                                                                                                        |                     |
| <b>Description</b> | Compile switch to enable and disable the Fls_BlankCheck function.                                                       |                     |
| <b>Screenshot</b>  | <a href="#">Fls Blank Check Api</a>  |                     |
| <b>Properties</b>  | <b>Property</b>                                                                                                         | <b>Value</b>        |
|                    | Type                                                                                                                    | Variable:Boolean    |
|                    | Label                                                                                                                   | Fls Blank Check Api |
|                    | Origin                                                                                                                  | AUTOSAR_ECUC        |
|                    | Default                                                                                                                 | false               |

### 3.2.2.4 FlsCancelApi

|                    |                                                                                                                    |                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Variables</b>   | FlsCancelApi                                                                                                       |                  |
| <b>Description</b> | Compile switch to enable and disable the Fls_Cancel function.                                                      |                  |
| <b>Screenshot</b>  | <a href="#">Fls Cancel Api</a>  |                  |
| <b>Properties</b>  | <b>Property</b>                                                                                                    | <b>Value</b>     |
|                    | Type                                                                                                               | Variable:Boolean |
|                    | Label                                                                                                              | Fls Cancel Api   |
|                    | Origin                                                                                                             | AUTOSAR_ECUC     |
|                    | Default                                                                                                            | false            |

### 3.2.2.5 FlsCompareApi

|                    |                                                                                                                       |                  |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Variables</b>   | FlsCompareApi                                                                                                         |                  |
| <b>Description</b> | Compile switch to enable and disable the Fls_Compare function.                                                        |                  |
| <b>Screenshot</b>  | <a href="#">Fls Compare Api</a>  |                  |
| <b>Properties</b>  | <b>Property</b>                                                                                                       | <b>Value</b>     |
|                    | Type                                                                                                                  | Variable:Boolean |
|                    | Label                                                                                                                 | Fls Compare Api  |

|  |         |              |
|--|---------|--------------|
|  | Origin  | AUTOSAR_ECUC |
|  | Default | false        |

### 3.2.2.6 FlsDevErrorDetect

|                    |                                                                                                                                                                                                                                                                                                                                                                                      |                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Variables</b>   | FlsCompareApi                                                                                                                                                                                                                                                                                                                                                                        |                              |
| <b>Description</b> | Switches the Default Error Tracer (Det) detection and notification ON or OFF.                                                                                                                                                                                                                                                                                                        |                              |
| <b>Screenshot</b>  | <a href="#">Fls Development Error Detect</a>     |                              |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                      | <b>Value</b>                 |
|                    | Type                                                                                                                                                                                                                                                                                                                                                                                 | Variable:Boolean             |
|                    | Label                                                                                                                                                                                                                                                                                                                                                                                | Fls Development Error Detect |
|                    | Origin                                                                                                                                                                                                                                                                                                                                                                               | AUTOSAR_ECUC                 |
|                    | Default                                                                                                                                                                                                                                                                                                                                                                              | false                        |

### 3.2.2.7 FlsDriverIndex

|                    |                                                                                                                                                                                                                                                                                                                                           |                  |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Variables</b>   | FlsDriverIndex                                                                                                                                                                                                                                                                                                                            |                  |
| <b>Description</b> | Index of the driver, used by FEE.                                                                                                                                                                                                                                                                                                         |                  |
| <b>Screenshot</b>  | <a href="#">Fls Driver Index (0 -&gt; 254)</a>  <input type="text" value="0"/>   |                  |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                                           | <b>Value</b>     |
|                    | Type                                                                                                                                                                                                                                                                                                                                      | Variable:Integer |
|                    | Label                                                                                                                                                                                                                                                                                                                                     | Fls Driver Index |
|                    | Origin                                                                                                                                                                                                                                                                                                                                    | AUTOSAR_ECUC     |
|                    | Default                                                                                                                                                                                                                                                                                                                                   | 0                |

### 3.2.2.8 FlsEraseVerificationEnabled

|                    |                                                                                                                                                                                                                                                                                                                                   |                        |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Variables</b>   | FlsEraseVerificationEnabled                                                                                                                                                                                                                                                                                                       |                        |
| <b>Description</b> | Compile switch to enable erase verification.                                                                                                                                                                                                                                                                                      |                        |
| <b>Screenshot</b>  | <a href="#">Fls Erase Verification</a>  <input type="checkbox"/>   |                        |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                                   | <b>Value</b>           |
|                    | Type                                                                                                                                                                                                                                                                                                                              | Variable:Boolean       |
|                    | Label                                                                                                                                                                                                                                                                                                                             | Fls Erase Verification |
|                    | Origin                                                                                                                                                                                                                                                                                                                            | AUTOSAR_ECUC           |
|                    | Default                                                                                                                                                                                                                                                                                                                           | false                  |

### 3.2.2.9 FlsGetJobResultApi

|                    |                                                                                                                                                                                                                                                                                                                                                                                        |                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Variables</b>   | FlsGetJobResultApi                                                                                                                                                                                                                                                                                                                                                                     |                        |
| <b>Description</b> | Compile switch to enable and disable the Fls_GetJobResult function.                                                                                                                                                                                                                                                                                                                    |                        |
| <b>Screenshot</b>  | <a href="#">Fls Get Job Result Api</a>     |                        |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Value</b>           |
|                    | Type                                                                                                                                                                                                                                                                                                                                                                                   | Variable:Boolean       |
|                    | Label                                                                                                                                                                                                                                                                                                                                                                                  | Fls Get Job Result Api |
|                    | Origin                                                                                                                                                                                                                                                                                                                                                                                 | AUTOSAR_ECUC           |

|  |         |       |
|--|---------|-------|
|  | Default | false |
|--|---------|-------|

### 3.2.2.10 FlsGetStatusApi

|                    |                                                                                                                                                                                                                                                                                                                                                                                    |                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Variables</b>   | FlsGetStatusApi                                                                                                                                                                                                                                                                                                                                                                    |                    |
| <b>Description</b> | Compile switch to enable and disable the Fls_GetStatus function.                                                                                                                                                                                                                                                                                                                   |                    |
| <b>Screenshot</b>  | <a href="#">Fls Get Status Api</a>     |                    |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                    | <b>Value</b>       |
|                    | Type                                                                                                                                                                                                                                                                                                                                                                               | Variable:Boolean   |
|                    | Label                                                                                                                                                                                                                                                                                                                                                                              | Fls Get Status Api |
|                    | Origin                                                                                                                                                                                                                                                                                                                                                                             | AUTOSAR_ECUC       |
|                    | Default                                                                                                                                                                                                                                                                                                                                                                            | false              |

### 3.2.2.11 FlsSetModeApi

|                    |                                                                                                                                                                                                                                                                                                                                                                          |                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Variables</b>   | FlsGetJobResultApi                                                                                                                                                                                                                                                                                                                                                       |                  |
| <b>Description</b> | Compile switch to enable and disable the Fls_SetMode function.                                                                                                                                                                                                                                                                                                           |                  |
| <b>Screenshot</b>  | <a href="#">Fls Set Mode Api</a>     |                  |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                          | <b>Value</b>     |
|                    | Type                                                                                                                                                                                                                                                                                                                                                                     | Variable:Boolean |
|                    | Label                                                                                                                                                                                                                                                                                                                                                                    | Fls Set Mode Api |
|                    | Origin                                                                                                                                                                                                                                                                                                                                                                   | AUTOSAR_ECUC     |
|                    | Default                                                                                                                                                                                                                                                                                                                                                                  | false            |

### 3.2.2.12 FlsTimeoutSupervisionEnabled

|                    |                                                                                                                                                                                                                                                                                                                                                                                                 |                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variables</b>   | FlsTimeoutSupervisionEnabled                                                                                                                                                                                                                                                                                                                                                                    |                         |
| <b>Description</b> | Compile switch to enable timeout supervision.                                                                                                                                                                                                                                                                                                                                                   |                         |
| <b>Screenshot</b>  | <a href="#">Fls Timeout Supervision</a>     |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                 | <b>Value</b>            |
|                    | Type                                                                                                                                                                                                                                                                                                                                                                                            | Variable:Boolean        |
|                    | Label                                                                                                                                                                                                                                                                                                                                                                                           | Fls Timeout Supervision |
|                    | Origin                                                                                                                                                                                                                                                                                                                                                                                          | AUTOSAR_ECUC            |
|                    | Default                                                                                                                                                                                                                                                                                                                                                                                         | false                   |

### 3.2.2.13 FlsTotalSize

|                    |                                                                                                                                                                                                                                                                                                                       |                  |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Variables</b>   | FlsTotalSize                                                                                                                                                                                                                                                                                                          |                  |
| <b>Description</b> | The total amount of flash memory in bytes.                                                                                                                                                                                                                                                                            |                  |
| <b>Screenshot</b>  | <a href="#">Fls Total Size (0 -&gt; 4294967295)</a>    |                  |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                                                                                                                                                                       | <b>Value</b>     |
|                    | Type                                                                                                                                                                                                                                                                                                                  | Variable:Integer |
|                    | Label                                                                                                                                                                                                                                                                                                                 | Fls Total Size   |
|                    | Origin                                                                                                                                                                                                                                                                                                                | AUTOSAR_ECUC     |
|                    | Default                                                                                                                                                                                                                                                                                                               | 0                |

## 3.2.2.14 FlsUseInterrupts

|                    |                                                                                     |                    |
|--------------------|-------------------------------------------------------------------------------------|--------------------|
| <b>Variables</b>   | FlsUseInterrupts                                                                    |                    |
| <b>Description</b> | Job processing triggered by hardware interrupt.                                     |                    |
| <b>Screenshot</b>  |  |                    |
| <b>Properties</b>  | <b>Property</b>                                                                     | <b>Value</b>       |
|                    | Type                                                                                | Variable:Boolean   |
|                    | Label                                                                               | Fls Use Interrupts |
|                    | Origin                                                                              | AUTOSAR_ECUC       |
|                    | Default                                                                             | false              |

## 3.2.2.15 FlsVersionInfoApi

|                    |                                                                                               |                      |
|--------------------|-----------------------------------------------------------------------------------------------|----------------------|
| <b>Variables</b>   | FlsVersionInfoApi                                                                             |                      |
| <b>Description</b> | Pre-processor switch to enable / disable the API to read out the modules version information. |                      |
| <b>Screenshot</b>  |              |                      |
| <b>Properties</b>  | <b>Property</b>                                                                               | <b>Value</b>         |
|                    | Type                                                                                          | Variable:Boolean     |
|                    | Label                                                                                         | Fls Version Info Api |
|                    | Origin                                                                                        | AUTOSAR_ECUC         |
|                    | Default                                                                                       | false                |

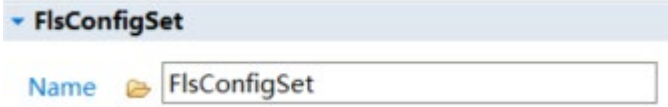
## 3.2.2.16 FlsWriteVerificationEnabled

|                    |                                                                                       |                        |
|--------------------|---------------------------------------------------------------------------------------|------------------------|
| <b>Variables</b>   | FlsWriteVerificationEnabled                                                           |                        |
| <b>Description</b> | Compile switch to enable write verification.                                          |                        |
| <b>Screenshot</b>  |  |                        |
| <b>Properties</b>  | <b>Property</b>                                                                       | <b>Value</b>           |
|                    | Type                                                                                  | Variable:Boolean       |
|                    | Label                                                                                 | Fls Write Verification |
|                    | Origin                                                                                | AUTOSAR_ECUC           |
|                    | Default                                                                               | false                  |

## 3.2.2.17 FlsEcucPartitionRef

|                    |                                                                                                                                  |                     |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Variables</b>   | FlsEcucPartitionRef                                                                                                              |                     |
| <b>Description</b> | Maps the Flash driver to zero or one ECUC partition to make the driver API available in this partition. Not used in driver code. |                     |
| <b>Screenshot</b>  |                                             |                     |
| <b>Properties</b>  | <b>Property</b>                                                                                                                  | <b>Value</b>        |
|                    | Type                                                                                                                             | Container           |
|                    | Label                                                                                                                            | FlsEcucPartitionRef |
|                    | Origin                                                                                                                           | AUTOSAR_ECUC        |
|                    | Default                                                                                                                          | N/A                 |

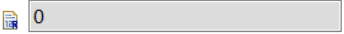
### 3.2.3 FlsConfigSet

|                    |                                                                                                |              |
|--------------------|------------------------------------------------------------------------------------------------|--------------|
| <b>Container</b>   | FlsConfigSet                                                                                   |              |
| <b>Description</b> | Container for general parameters of the flash driver. These parameters are always pre-compile. |              |
| <b>Screenshot</b>  |               |              |
| <b>Properties</b>  | <b>Property</b>                                                                                | <b>Value</b> |
|                    | Type                                                                                           | Container    |

#### 3.2.3.1 FlsAcErase

|                    |                                                                                                                                             |                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Variables</b>   | FlsAcErase                                                                                                                                  |                       |
| <b>Description</b> | Address offset in RAM to which the erase flash access code shall be loaded. Used as function pointer to access the erase flash access code. |                       |
| <b>Screenshot</b>  |                                                          |                       |
| <b>Properties</b>  | <b>Property</b>                                                                                                                             | <b>Value</b>          |
|                    | Type                                                                                                                                        | Variable: Integer     |
|                    | Label                                                                                                                                       | Fls Access Code Erase |
|                    | Origin                                                                                                                                      | AUTOSAR_ECUC          |
|                    | Default                                                                                                                                     | 0                     |

#### 3.2.3.2 FlsAcWrite

|                    |                                                                                                                                             |                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Variables</b>   | FlsAcWrite                                                                                                                                  |                       |
| <b>Description</b> | Address offset in RAM to which the write flash access code shall be loaded. Used as function pointer to access the write flash access code. |                       |
| <b>Screenshot</b>  |                                                        |                       |
| <b>Properties</b>  | <b>Property</b>                                                                                                                             | <b>Value</b>          |
|                    | Type                                                                                                                                        | Variable: Integer     |
|                    | Label                                                                                                                                       | Fls Access Code Write |
|                    | Origin                                                                                                                                      | AUTOSAR_ECUC          |
|                    | Default                                                                                                                                     | 0                     |

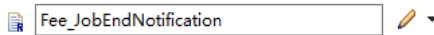
#### 3.2.3.3 FlsCallCycle

|                    |                                                                                       |                |
|--------------------|---------------------------------------------------------------------------------------|----------------|
| <b>Variables</b>   | FlsCallCycle                                                                          |                |
| <b>Description</b> | Cycle time of calls of the flash driver main function                                 |                |
| <b>Screenshot</b>  |  |                |
| <b>Properties</b>  | <b>Property</b>                                                                       | <b>Value</b>   |
|                    | Type                                                                                  | Variable:Float |
|                    | Label                                                                                 | Fls Call Cycle |
|                    | Origin                                                                                | AUTOSAR_ECUC   |
|                    | Default                                                                               | 0.0            |

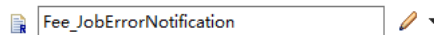
## 3.2.3.4 FlsDefaultMode

|                    |                                                                                                                                                                       |                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Variables</b>   | FlsDefaultMode                                                                                                                                                        |                       |
| <b>Description</b> | This parameter is the default FLS device mode after initialization.                                                                                                   |                       |
| <b>Screenshot</b>  |   |                       |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                       | <b>Value</b>          |
|                    | Type                                                                                                                                                                  | Variable: enumeration |
|                    | Label                                                                                                                                                                 | Fls Default Mode      |
|                    | Origin                                                                                                                                                                | AUTOSAR_ECUC          |
|                    | Default                                                                                                                                                               | MEMIF_MODE_SLOW       |

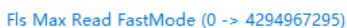
## 3.2.3.5 FlsJobEndNotification

|                    |                                                                                                                                                                       |                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Variables</b>   | FlsJobEndNotification                                                                                                                                                 |                          |
| <b>Description</b> | Mapped to the job end notification routine provided by some upper layer module, typically the Fee module.                                                             |                          |
| <b>Screenshot</b>  |   |                          |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                       | <b>Value</b>             |
|                    | Type                                                                                                                                                                  | Variable: String         |
|                    | Label                                                                                                                                                                 | Fls Job End Notification |
|                    | Origin                                                                                                                                                                | AUTOSAR_ECUC             |
|                    | Default                                                                                                                                                               | Fee_JobEndNotification   |

## 3.2.3.6 FlsJobErrorNotification

|                    |                                                                                                                                                                           |                            |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Variables</b>   | FlsJobErrorNotification                                                                                                                                                   |                            |
| <b>Description</b> | Mapped to the job error notification routine provided by some upper layer module, typically the Fee module.                                                               |                            |
| <b>Screenshot</b>  |   |                            |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                           | <b>Value</b>               |
|                    | Type                                                                                                                                                                      | Variable: String           |
|                    | Label                                                                                                                                                                     | Fls Job Error Notification |
|                    | Origin                                                                                                                                                                    | AUTOSAR_ECUC               |
|                    | Default                                                                                                                                                                   | Fee_JobErrorNotification   |

## 3.2.3.7 FlsMaxReadFastMode

|                    |                                                                                                                                                                           |                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Variables</b>   | FlsMaxReadFastMode                                                                                                                                                        |                       |
| <b>Description</b> | The maximum number of bytes to read or compare in one cycle of the flash driver's job processing function in fast mode.                                                   |                       |
| <b>Screenshot</b>  |   |                       |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                                           | <b>Value</b>          |
|                    | Type                                                                                                                                                                      | Variable: Integer     |
|                    | Label                                                                                                                                                                     | Fls Max Read FastMode |
|                    | Origin                                                                                                                                                                    | AUTOSAR_ECUC          |
|                    | Default                                                                                                                                                                   | 1048576               |



## 3.2.3.8 FlsMaxReadNormalMode

|                    |                                                                                                                                                   |                          |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Variables</b>   | FlsMaxReadNormalMode                                                                                                                              |                          |
| <b>Description</b> | The maximum number of bytes to read or compare in one cycle of the flash driver's job processing function in normal mode.                         |                          |
| <b>Screenshot</b>  | <a href="#">Fls Max Read Normal Mode (0 -&gt; 4294967295)</a>  |                          |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                   | <b>Value</b>             |
|                    | Type                                                                                                                                              | Variable: Integer        |
|                    | Label                                                                                                                                             | Fls Max Read Normal Mode |
|                    | Origin                                                                                                                                            | AUTOSAR_ECUC             |
|                    | Default                                                                                                                                           | 1024                     |

## 3.2.3.9 FlsMaxWriteFastMode

|                    |                                                                                                                             |                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variables</b>   | FlsMaxWriteFastMode                                                                                                         |                         |
| <b>Description</b> | The maximum number of bytes to write in one cycle of the flash driver's job processing function in fast mode.               |                         |
| <b>Screenshot</b>  | <a href="#">Fls Max Write Fast Mode</a>  |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                                             | <b>Value</b>            |
|                    | Type                                                                                                                        | Variable: Integer       |
|                    | Label                                                                                                                       | Fls Max Write Fast Mode |
|                    | Origin                                                                                                                      | AUTOSAR_ECUC            |
|                    | Default                                                                                                                     | 128                     |

## 3.2.3.10 FlsMaxWriteNormalMode

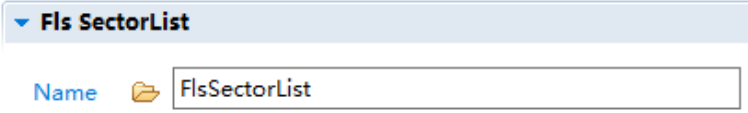
|                    |                                                                                                                                 |                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Variables</b>   | FlsMaxWriteNormalMode                                                                                                           |                           |
| <b>Description</b> | The maximum number of bytes to write in one cycle of the flash driver's job processing function in normal mode.                 |                           |
| <b>Screenshot</b>  | <a href="#">Fls Max Write Normal Mode</a>  |                           |
| <b>Properties</b>  | <b>Property</b>                                                                                                                 | <b>Value</b>              |
|                    | Type                                                                                                                            | Variable: Integer         |
|                    | Label                                                                                                                           | Fls Max Write Normal Mode |
|                    | Origin                                                                                                                          | AUTOSAR_ECUC              |
|                    | Default                                                                                                                         | 16                        |

## 3.2.3.11 FlsProtection

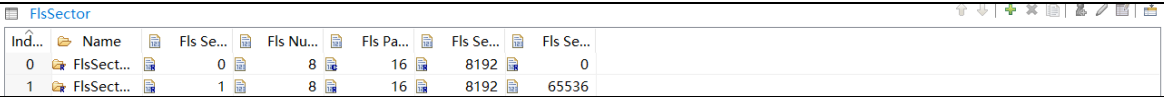
|                    |                                                                                                                                           |                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Variables</b>   | FlsMaxWriteNormalMode                                                                                                                     |                   |
| <b>Description</b> | Erase/write protection settings. Only relevant if supported by hardware.                                                                  |                   |
| <b>Screenshot</b>  | <a href="#">Fls Protection (0 -&gt; 4294967295)</a>  |                   |
| <b>Properties</b>  | <b>Property</b>                                                                                                                           | <b>Value</b>      |
|                    | Type                                                                                                                                      | Variable: Integer |
|                    | Label                                                                                                                                     | Fls Protection    |

|  |         |              |
|--|---------|--------------|
|  | Origin  | AUTOSAR_ECUC |
|  | Default | 1            |

### 3.2.4 FlsSectorList

|                    |                                                                                    |              |
|--------------------|------------------------------------------------------------------------------------|--------------|
| <b>Container</b>   | FlsSectorList                                                                      |              |
| <b>Description</b> | List of flashable sectors and pages.                                               |              |
| <b>Screenshot</b>  |  |              |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b> |
|                    | Type                                                                               | Container    |

#### 3.2.4.1 FlsSector

|                    |                                                                                    |              |
|--------------------|------------------------------------------------------------------------------------|--------------|
| <b>List</b>        | FlsSector                                                                          |              |
| <b>Description</b> | Erase/write protection settings. Only relevant if supported by hardware.           |              |
| <b>Screenshot</b>  |  |              |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b> |
|                    | Type                                                                               | list         |
|                    | Label                                                                              | FlsSector    |

#### 3.2.4.1.1 Fls Sector Index

|                    |                                                                                                                          |                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Variable</b>    | Fls Sector Index                                                                                                         |                  |
| <b>Description</b> | Fls Sector Index is an invariant index, used to order flash sectors and loop over them in the correct, configured order. |                  |
| <b>Screenshot</b>  |                                      |                  |
| <b>Properties</b>  | <b>Property</b>                                                                                                          | <b>Value</b>     |
|                    | Type                                                                                                                     | Variable:Integer |
|                    | Label                                                                                                                    | Fls Sector Index |
|                    | Origin                                                                                                                   | FLAGCHIP         |
|                    | Editable                                                                                                                 | False            |

#### 3.2.4.1.2 FlsNumberOfSectors

|                    |                                                                                      |                      |
|--------------------|--------------------------------------------------------------------------------------|----------------------|
| <b>Variable</b>    | FlsNumberOfSectors                                                                   |                      |
| <b>Description</b> | The number of specified PortPins in this PortContainer.                              |                      |
| <b>Screenshot</b>  |  |                      |
| <b>Properties</b>  | <b>Property</b>                                                                      | <b>Value</b>         |
|                    | Type                                                                                 | Variable: Integer    |
|                    | Label                                                                                | Fls Number Of Sector |
|                    | Origin                                                                               | AUTOSAR_ECUC         |
|                    | Editable                                                                             | True                 |
|                    | Default                                                                              | 1                    |

## 3.2.4.1.3 FlsPageSize

|                    |                                                                                                                                        |                   |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>List</b>        | FlsPageSize                                                                                                                            |                   |
| <b>Description</b> | Size of one page of this sector.                                                                                                       |                   |
| <b>Screenshot</b>  | <a href="#">Fls Page Size (0 -&gt; 4294967295)</a>  |                   |
| <b>Properties</b>  | <b>Property</b>                                                                                                                        | <b>Value</b>      |
|                    | Type                                                                                                                                   | Variable: Integer |
|                    | Label                                                                                                                                  | Fls Page Size     |
|                    | Origin                                                                                                                                 | AUTOSAR_ECUC      |
|                    | Editable                                                                                                                               | False             |
|                    | Default                                                                                                                                | 16                |

## 3.2.4.1.4 FlsSectorSize

|                    |                                                                                                                                          |                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Container</b>   | FlsSectorSize                                                                                                                            |                 |
| <b>Description</b> | Size of this sector.                                                                                                                     |                 |
| <b>Screenshot</b>  | <a href="#">Fls Sector Size (0 -&gt; 4294967295)</a>  |                 |
| <b>Properties</b>  | <b>Property</b>                                                                                                                          | <b>Value</b>    |
|                    | Type                                                                                                                                     | Container       |
|                    | Label                                                                                                                                    | Fls Sector Size |
|                    | Origin                                                                                                                                   | AUTOSAR_ECUC    |
|                    | Editable                                                                                                                                 | False           |
|                    | Default                                                                                                                                  | 8192            |

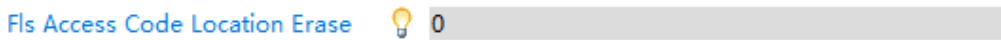
## 3.2.4.1.5 FlsSectorStartaddress

|                    |                                                                                                                                                     |                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Variable</b>    | FlsSectorStartaddress                                                                                                                               |                          |
| <b>Description</b> | Start address of this sector.                                                                                                                       |                          |
| <b>Screenshot</b>  | <a href="#">Fls Sector Start Address (0 -&gt; 4294967295)</a>  |                          |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                     | <b>Value</b>             |
|                    | Type                                                                                                                                                | Variable: Integer        |
|                    | Label                                                                                                                                               | Fls Sector Start Address |
|                    | Editable                                                                                                                                            | True                     |
|                    | Default                                                                                                                                             | 0                        |

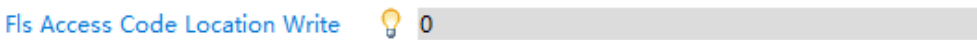
## 3.2.5 FlsPublishedInformation

|                    |                                                                                                                                                               |              |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Container</b>   | FlsPublishedInformation                                                                                                                                       |              |
| <b>Description</b> | Additional published parameters not covered by CommonPublishedInformation container.                                                                          |              |
| <b>Screenshot</b>  | <a href="#">Name</a>  <input type="text" value="FlsPublishedInformation"/> |              |
| <b>Properties</b>  | <b>Property</b>                                                                                                                                               | <b>Value</b> |
|                    | Type                                                                                                                                                          | Container    |

## 3.2.5.1 FlsAcLocationErase

|                    |                                                                                    |                                |
|--------------------|------------------------------------------------------------------------------------|--------------------------------|
| <b>Variable</b>    | FlsAcLocationErase                                                                 |                                |
| <b>Description</b> | Position in RAM, to which the erase flash access code has to be loaded.            |                                |
| <b>Screenshot</b>  |  |                                |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b>                   |
|                    | Type                                                                               | Variable: Integer              |
|                    | Label                                                                              | Fls Access Code Location Erase |
|                    | Origin                                                                             | AUTOSAR_ECUC                   |
|                    | Default                                                                            | 0                              |

## 3.2.5.2 FlsAcLocationWrite

|                    |                                                                                    |                                |
|--------------------|------------------------------------------------------------------------------------|--------------------------------|
| <b>Variable</b>    | FlsAcLocationWrite                                                                 |                                |
| <b>Description</b> | Position in RAM, to which the write flash access code has to be loaded.            |                                |
| <b>Screenshot</b>  |  |                                |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b>                   |
|                    | Type                                                                               | Variable: Integer              |
|                    | Label                                                                              | Fls Access Code Location Write |
|                    | Origin                                                                             | AUTOSAR_ECUC                   |
|                    | Default                                                                            | 0                              |

## 3.2.5.2.1 FlsAcSizeErase

|                    |                                                                                      |                            |
|--------------------|--------------------------------------------------------------------------------------|----------------------------|
| <b>Variable</b>    | FlsAcSizeErase                                                                       |                            |
| <b>Description</b> | Number of bytes in RAM needed for the erase flash access code.                       |                            |
| <b>Screenshot</b>  |  |                            |
| <b>Properties</b>  | <b>Property</b>                                                                      | <b>Value</b>               |
|                    | Type                                                                                 | Variable: Integer          |
|                    | Label                                                                                | Fls Access Code Size Erase |
|                    | Origin                                                                               | AUTOSAR_ECUC               |
|                    | Default                                                                              | 0                          |

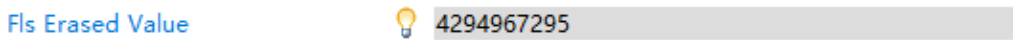
## 3.2.5.3 FlsAcSizeWrite

|                    |                                                                                      |                            |
|--------------------|--------------------------------------------------------------------------------------|----------------------------|
| <b>Variable</b>    | FlsAcSizeWrite                                                                       |                            |
| <b>Description</b> | Number of bytes in RAM needed for the write flash access code.                       |                            |
| <b>Screenshot</b>  |  |                            |
| <b>Properties</b>  | <b>Property</b>                                                                      | <b>Value</b>               |
|                    | Type                                                                                 | Variable: Integer          |
|                    | Label                                                                                | Fls Access Code Size Write |
|                    | Origin                                                                               | AUTOSAR_ECUC               |
|                    | Default                                                                              | 0                          |

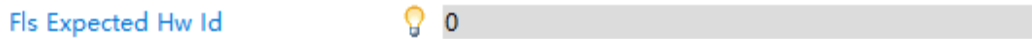
## 3.2.5.4 FlsEraseTime

|                    |                                                                                    |                 |
|--------------------|------------------------------------------------------------------------------------|-----------------|
| <b>Variable</b>    | FlsEraseTime                                                                       |                 |
| <b>Description</b> | Maximum time to erase one complete flash sector [sec].                             |                 |
| <b>Screenshot</b>  |  |                 |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b>    |
|                    | Type                                                                               | Variable: Float |
|                    | Label                                                                              | Fls Erase Time  |
|                    | Origin                                                                             | AUTOSAR_ECUC    |
|                    | Default                                                                            | 0.0032          |

## 3.2.5.5 FlsErasedValue

|                    |                                                                                    |                      |
|--------------------|------------------------------------------------------------------------------------|----------------------|
| <b>Variable</b>    | FlsErasedValue                                                                     |                      |
| <b>Description</b> | The contents of an erased flash memory cell.                                       |                      |
| <b>Screenshot</b>  |  |                      |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b>         |
|                    | Type                                                                               | Fls Erased Value     |
|                    | Label                                                                              | Port VersionInfo Api |
|                    | Origin                                                                             | AUTOSAR_ECUC         |
|                    | Default                                                                            | 4294967295           |

## 3.2.5.6 FlsExpectedHwId

|                    |                                                                                      |                    |
|--------------------|--------------------------------------------------------------------------------------|--------------------|
| <b>Variable</b>    | FlsExpectedHwId                                                                      |                    |
| <b>Description</b> | Unique identifier of the hardware device that is expected by this driver.            |                    |
| <b>Screenshot</b>  |  |                    |
| <b>Properties</b>  | <b>Property</b>                                                                      | <b>Value</b>       |
|                    | Type                                                                                 | Variable: String   |
|                    | Label                                                                                | Fls Expected Hw Id |
|                    | Origin                                                                               | AUTOSAR_ECUC       |
|                    | Default                                                                              | 0                  |

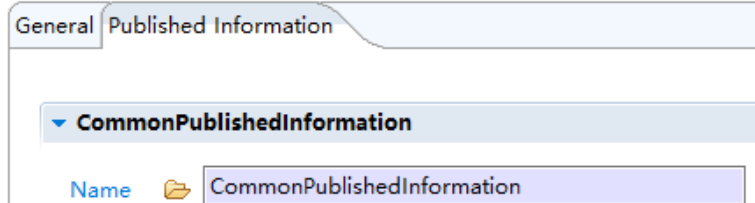
## 3.2.5.7 FlsSpecifiedEraseCycles

|                    |                                                                                      |                            |
|--------------------|--------------------------------------------------------------------------------------|----------------------------|
| <b>Variable</b>    | FlsSpecifiedEraseCycles                                                              |                            |
| <b>Description</b> | Number of erase cycles specified for the flash device.                               |                            |
| <b>Screenshot</b>  |  |                            |
| <b>Properties</b>  | <b>Property</b>                                                                      | <b>Value</b>               |
|                    | Type                                                                                 | Variable: Integer          |
|                    | Label                                                                                | Fls Specified Erase Cycles |
|                    | Origin                                                                               | AUTOSAR_ECUC               |
|                    | Default                                                                              | 100000                     |

## 3.2.5.8 FlsWriteTime

|                    |                                                                                    |                 |
|--------------------|------------------------------------------------------------------------------------|-----------------|
| <b>Variable</b>    | FlsWriteTime                                                                       |                 |
| <b>Description</b> | Maximum time to program one complete flash page [sec].                             |                 |
| <b>Screenshot</b>  |  |                 |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b>    |
|                    | Type                                                                               | Variable: Float |
|                    | Label                                                                              | Fls Write Time  |
|                    | Origin                                                                             | AUTOSAR_ECUC    |
|                    | Default                                                                            | 1.2E-4          |

## 3.2.6 CommonPublishedInformation

|                    |                                                                                                           |  |
|--------------------|-----------------------------------------------------------------------------------------------------------|--|
| <b>Container</b>   | CommonPublishedInformation                                                                                |  |
| <b>Description</b> | Common container, aggregated by all modules. It contains published information about vendor and versions. |  |
| <b>Screenshot</b>  |                         |  |
| <b>Properties</b>  | N/A                                                                                                       |  |

## 3.2.6.1 ArReleaseMajorVersion

|                    |                                                                                                 |                         |
|--------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | ArReleaseMajorVersion                                                                           |                         |
| <b>Description</b> | Major version number of AUTOSAR specification on which the appropriate implementation is based. |                         |
| <b>Screenshot</b>  |             |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                 | <b>Value</b>            |
|                    | Type                                                                                            | Variable: Integer_Label |
|                    | Origin                                                                                          | FLAGCHIP                |
|                    | SymbolicNameValue                                                                               | false                   |
|                    | Default                                                                                         | 4                       |

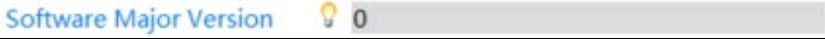
## 3.2.6.2 ArReleaseMinorVersion

|                    |                                                                                                 |                         |
|--------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | ArReleaseMinorVersion                                                                           |                         |
| <b>Description</b> | Minor version number of AUTOSAR specification on which the appropriate implementation is based. |                         |
| <b>Screenshot</b>  |             |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                 | <b>Value</b>            |
|                    | Type                                                                                            | Variable: Integer_Label |
|                    | Origin                                                                                          | FLAGCHIP                |
|                    | SymbolicNameValue                                                                               | False                   |
|                    | Default                                                                                         | 6                       |

## 3.2.6.3 ArReleaseRevisionVersion

|                    |                                                                                                    |                         |
|--------------------|----------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | ArReleaseRevisionVersion                                                                           |                         |
| <b>Description</b> | Revision version number of AUTOSAR specification on which the appropriate implementation is based. |                         |
| <b>Screenshot</b>  |                  |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                    | <b>Value</b>            |
|                    | Type                                                                                               | Variable: Integer_Label |
|                    | Origin                                                                                             | FLAGCHIP                |
|                    | SymbolicNameValue                                                                                  | False                   |
|                    | Default                                                                                            | 0                       |

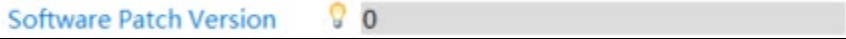
## 3.2.6.4 SwMajorVersion

|                    |                                                                                                             |                         |
|--------------------|-------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | SwMajorVersion                                                                                              |                         |
| <b>Description</b> | Major version number of the vendor specific implementation of the module. The numbering is vendor specific. |                         |
| <b>Screenshot</b>  |                           |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                             | <b>Value</b>            |
|                    | Type                                                                                                        | Variable: Integer_Label |
|                    | Origin                                                                                                      | FLAGCHIP                |
|                    | SymbolicNameValue                                                                                           | False                   |
|                    | Default                                                                                                     | 0                       |

## 3.2.6.5 SwMinorVersion

|                    |                                                                                                             |                         |
|--------------------|-------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | SwMinorVersion                                                                                              |                         |
| <b>Description</b> | Minor version number of the vendor specific implementation of the module. The numbering is vendor specific. |                         |
| <b>Screenshot</b>  |                         |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                             | <b>Value</b>            |
|                    | Type                                                                                                        | Variable: Integer_Label |
|                    | Origin                                                                                                      | FLAGCHIP                |
|                    | SymbolicNameValue                                                                                           | False                   |
|                    | Default                                                                                                     | 3                       |

## 3.2.6.6 SwPatchVersion

|                    |                                                                                                                   |                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | SwPatchVersion                                                                                                    |                         |
| <b>Description</b> | Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific. |                         |
| <b>Screenshot</b>  |                               |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                                   | <b>Value</b>            |
|                    | Type                                                                                                              | Variable: Integer_Label |
|                    | Origin                                                                                                            | FLAGCHIP                |
|                    | SymbolicNameValue                                                                                                 | False                   |
|                    | Default                                                                                                           | 0                       |

## 3.2.6.7 ModuleId

|                    |                                                                                    |                         |
|--------------------|------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | ModuleId                                                                           |                         |
| <b>Description</b> | Module ID of this module from Module List.                                         |                         |
| <b>Screenshot</b>  |  |                         |
| <b>Properties</b>  | <b>Property</b>                                                                    | <b>Value</b>            |
|                    | Type                                                                               | Variable: Integer_Label |
|                    | Origin                                                                             | FLAGCHIP                |
|                    | SymbolicNameValue                                                                  | False                   |
|                    | Default                                                                            | 92                      |

## 3.2.6.8 VendorId

|                    |                                                                                                |                         |
|--------------------|------------------------------------------------------------------------------------------------|-------------------------|
| <b>Variable</b>    | VendorId                                                                                       |                         |
| <b>Description</b> | Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list. |                         |
| <b>Screenshot</b>  |              |                         |
| <b>Properties</b>  | <b>Property</b>                                                                                | <b>Value</b>            |
|                    | Type                                                                                           | Variable: Integer_Label |
|                    | Origin                                                                                         | FLAGCHIP                |
|                    | SymbolicNameValue                                                                              | False                   |
|                    | Default                                                                                        | 174                     |



# Chapter 4 Configuration Guides

## 4.1 Configuration Item Constraint

- 1) In the FlsGeneral container, the Fls Base Address will match the response value according to the CommonSubdivisive option of the Common module. If the value does not match, an error will be reported. You need to click Calculate Value to match again.

**Fls General**

Name FlsGeneral

Fls Load Access Code On Job Start ☐

Fls Base Address 26214

Fls Blank Check Api ☐ Fls Cancel Api ☒

Fls Compare Api ☐ Fls Development Error Detect ☒

Fls Driver Index (0 -> 254) 0

Fls Erase Verification ☐ Fls Get Job Result Api ☒

- 2) If the Fee module is used, in the FlsConfigSet container, the FlsMaxWriteNormalMode must be no less than the FeeVirtualPageSize in the FeeGeneral container in the Fee module, or an error will be reported.

**Fee**

Name Fee

General **FeeClusterGroup** FeeBlockConfiguration Published Information

**Fee General**

Name FeeGeneral

Fee Development Error Detect ☐

FeeMainFunctionPeriod (0.0000001 -> 100000) 0.005

Fee Nvm Job End Notification NvM\_JobEndNotification

Fee Nvm Job Error Notification NvM\_JobErrorNotification

Fee Polling Mode ☐ Fee Set Mode Supported ☐

Fee Version Info Api ☐

**Fee Virtual Page Size** 32

Fee Data Buffer Size 96

## FLS

Name FLS

General FLSector Published Information

▼ FLSConfigSet

Name FLSConfigSet

|                                            |                          |
|--------------------------------------------|--------------------------|
| FLS Access Code Erase (0 -> 4294967295)    | 0                        |
| FLS Access Code Write (0 -> 4294967295)    | 0                        |
| FLS Call Cycle (0 -> 1)                    | 0.0                      |
| FLS Default Mode                           | MEMIF_MODE_SLOW          |
| FLS Job End Notification                   | Fee_JobEndNotification   |
| FLS Job Error Notification                 | Fee_JobErrorNotification |
| FLS Max Read FastMode (0 -> 4294967295)    | 1048576                  |
| FLS Max Read Normal Mode (0 -> 4294967295) | 1024                     |
| FLS Max Write Fast Mode                    | 120                      |
| FLS Max Write Normal Mode*                 | 16                       |
| FLS Protection (0 -> 4294967295)           | 0                        |

## 4.2 FLS Usage Common Steps

Basically, the FLS module can be configured by following the below 4 steps:

- 1) Choose which API will be used.
- 2) Configure response options or parameters according to flash performance and requirements
- 3) Generate configuration files.

## 4.3 Demo Configuration (FC7300F8M)

▼ FLS General

Name FLSGeneral

|                                   |                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------|
| FLS Load Access Code On Job Start | <input type="checkbox"/>                                                                  |
| FLS Base Address                  | 67108864                                                                                  |
| FLS Blank Check Api               | <input type="checkbox"/> FLS Cancel Api <input checked="" type="checkbox"/>               |
| FLS Compare Api                   | <input type="checkbox"/> FLS Development Error Detect <input checked="" type="checkbox"/> |
| FLS Driver Index (0 -> 254)       | 0                                                                                         |
| FLS Erase Verification            | <input type="checkbox"/> FLS Get Job Result Api <input checked="" type="checkbox"/>       |
| FLS Get Status Api                | <input checked="" type="checkbox"/> FLS Set Mode Api <input checked="" type="checkbox"/>  |
| FLS Timeout Supervision           | <input type="checkbox"/>                                                                  |
| FLS Total Size (0 -> 4294967295)  | 0                                                                                         |
| FLS Use Interrupts                | <input type="checkbox"/> FLS Version Info Api <input type="checkbox"/>                    |
| FLS Write Verification            | <input type="checkbox"/>                                                                  |

### FlsConfigSet

Name  FlsConfigSet

|                                                                                                              |                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fls Access Code Erase (0 -> 4294967295)                                                                      |  0                                                                                                            |
| Fls Access Code Write (0 -> 4294967295)                                                                      |  0                                                                                                            |
| Fls Call Cycle (0 -> 1)                                                                                      |  0.0                       |
| Fls Default Mode                                                                                             |  MEMIF_MODE_SLOW           |
|  Fls Job End Notification   |  Fee_JobEndNotification    |
|  Fls Job Error Notification |  Fee_JobErrorNotification  |
| Fls Max Read FastMode (0 -> 4294967295)                                                                      |  1048576                   |
| Fls Max Read Normal Mode (0 -> 4294967295)                                                                   |  1024                      |
| Fls Max Write Fast Mode                                                                                      |  128                       |
| Fls Max Write Normal Mode                                                                                    |  16                        |
| Fls Protection (0 -> 4294967295)                                                                             |  1                         |

### FlsSector

Name  FlsSector\_0

#### Fls Sector

|                                            |                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fls Sector Index                           |  0     |
| Fls Number Of Sector (0 -> 65535)          |  8     |
| Fls Page Size                              |  16    |
| Fls Sector Size                            |  8192  |
| Fls Sector Start Address (0 -> 4294967295) |  0     |

### FlsSector

Name  FlsSector\_1

#### Fls Sector

|                                            |                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fls Sector Index                           |  1      |
| Fls Number Of Sector (0 -> 65535)          |  8      |
| Fls Page Size                              |  16     |
| Fls Sector Size                            |  8192   |
| Fls Sector Start Address (0 -> 4294967295) |  65536  |